

West, Topologies & Hello World

Zephyr RTOS Course

The logo for iomico, featuring the word "iomico" in a lowercase, sans-serif font. Each letter is a different color: 'i' is yellow, 'o' is red, 'm' is green, 'i' is blue, 'c' is green, and 'o' is blue. The logo is set against a white, rounded rectangular background.



Quick Recap — L01



- What Zephyr is: RTOS by Linux Foundation, multi-arch, modular
- Zephyr vs FreeRTOS: different trade-offs, not better/worse
- What's inside: drivers, networking, 3rd party, subsystems
- Environment setup: dependencies, `venv`, `west`, SDK
- Built Hello World from `zephyr/samples/hello_world`



Format



- Lecture with optional **live demo**
- Homework after each session (checked by us)
- Recordings will be shared
- Questions addressed at the end in a **Q&A session**



Today's Agenda



- Meet `west` — Zephyr's meta-tool
- **Project topologies** — freestanding, repository, workspace
- The `west.yml` manifest
- Creating a **workspace from scratch**
- **Hello World** walkthrough — code, build, flash
- Troubleshooting tips
- Tasks



Meet west



Zephyr's Meta-Tool



- **Python-based** build orchestrator
- Handles:
 - **Repository management** (multi-module workspaces)
 - **Build system** integration (CMake + Ninja)
 - **Flashing**, debugging, logging
- Wraps `CMake` + `ninja` for convenience
- Extensible with custom commands



Key Subcommands



Command	Purpose
<code>west init</code>	Initialize a workspace
<code>west update</code>	Fetch/update all manifest dependencies
<code>west build</code>	Configure and build an application
<code>west flash</code>	Flash the built image to a board
<code>west debug</code>	Launch a debugger session
<code>west boards</code>	List all supported boards
<code>west zephyr-export</code>	Register Zephyr CMake package



Project Topologies



Freestanding Application



```
<home>/
├─ zephyrproject/
│   ├── .west/
│   │   └─ config
│   ├── zephyr/
│   └─ ...
└─ my-app/
    ├── CMakeLists.txt
    ├── prj.conf
    └─ src/
        └─ main.c
```



Repository Application



```
zephyrproject/  
├── .west/  
│   └── config  
└── zephyr/  
    ├── arch/  
    ├── boards/  
    ├── cmake/  
    ├── samples/  
    │   ├── hello_world/  
    │   └── ...  
    ├── tests/  
    └── ...
```



Workspace Application



```
my-project/  
├── .west/  
│   └── config  
├── app/  
│   ├── CMakeLists.txt  
│   ├── prj.conf  
│   ├── west.yml  
│   └── src/  
│       └── main.c  
├── zephyr/  
├── bootloader/  
├── modules/  
└── <vendor/private-repos>/
```



Which Topology to Use?



Topology	When to use
Freestanding	Quick experiments, learning, simple projects
Repository	Contributing to Zephyr samples/tests
Workspace	Production projects, custom boards, multiple apps

Recommendation: Use **Workspace** for real projects



The `west.yml` Manifest



west.yml



```
manifest:
  version: 1.0                                # manifest schema version
  projects:
    - name: zephyr                            # dependency name
      url: https://github.com/zephyrproject-rtos/zephyr
      revision: v4.2.0                        # pinned git tag
      import:
        name-allowlist:                       # only fetch what we need
          - cmsis
          - hal_nxp
        path-prefix: deps                    # put deps in deps/
```

- Omit `name-allowlist` to fetch all Zephyr sub-projects (~80 repos)



Creating a Workspace



Step 1: Initialize



```
# From a local directory
git clone https://github.com/iomico-public/zephyr-course
cd zephyr-course
west init -l

# Or from a remote repository
west init -m https://github.com/iomico-public/zephyr-course
```



Step 2: Fetch Dependencies



```
west update
```

This downloads:

- Zephyr (at the pinned revision)
- All allowlisted sub-projects (cmsis, hal_nxp, etc.)



Step 3: Set Up Environment



```
# Register Zephyr CMake package  
west zephyr-export
```



Resulting Structure



```
workspace/
├── .west/
│   └── config
├── zephyr-course/
│   ├── west.yml
│   └── app
│       ├── CMakeLists.txt
│       ├── prj.conf
│       └── src/
│           └── main.c
└── deps/
    ├── zephyr/
    ├── modules/
    │   ├── hal/
    │   └── nxp/
    └── ...
```



Hello World Walkthrough

12 directories, 2 files

```
~/p/o/io/i/demo ✓ < 19:40:46
source .zephyr/bin/activate

~/p/o/io/i/demo ✓ < .zephyr ✚ < 19:41:10
ls

deps zephyr-course zephyrproject

~/p/o/io/i/demo ✓ < .zephyr ✚ < 19:42:09
cd zephyr-course

~/p/o/io/i/d/zephyr-course > main ✓ < .zephyr ✚ < 19:42:19
ls

app build README.md west.yml

~/p/o/io/i/d/zephyr-course > main ✓ < .zephyr ✚ < 19:42:19
west init -l
FATAL ERROR: already initialized in /home/dms/projects/own/iomico/internal/demo, aborting.
Hint: if you do not want a workspace there,
remove this directory and re-run this command:

/home/dms/projects/own/iomico/internal/demo/.west

~/p/o/io/i/d/zephyr-course > main ! ✘ < .zephyr ✚ < 19:42:35
west update
=== updating zephyr (deps/zephyr):
HEAD is now at 413b789deb3 release: Zephyr v4.2.0
=== updating cmsis_6 (deps/modules/hal/cmsis_6):
HEAD is now at 06d952b6 IAR LDR{,B,H}T fix (#242)
=== updating hal_nordic (deps/modules/hal/nordic):
HEAD is now at 9587b1d nrfx: cracen: align to nrf54lm20a
=== updating hal_nxp (deps/modules/hal/nxp):
HEAD is now at 7a52cbb7c s32: s32ze: patch PSI5 for integration with zephyr
=== updating hal_stm32 (deps/modules/hal/stm32):
HEAD is now at 1e753266 license: Update Synopsys license terms in the license file and code

~/p/o/io/i/d/zephyr-course > main ✓ < .zephyr ✚ < 19:43:08
```

```
~/p/o/io/i/d/zephyr-course > main
west init -l
FATAL ERROR: already initialized in /home/dms/projects/own/iomico/internal/demo, aborting.
Hint: if you do not want a workspace there,
remove this directory and re-run this command:

/home/dms/projects/own/iomico/internal/demo/.west

~/p/o/io/i/d/zephyr-course > main
west update
=== updating zephyr (deps/zephyr):
HEAD is now at 413b789deb3 release: Zephyr v4.2.0
=== updating cmsis_6 (deps/modules/hal/cmsis_6):
HEAD is now at 06d952b6 IAR LDR{,B,H}T fix (#242)
=== updating hal_nordic (deps/modules/hal/nordic):
HEAD is now at 9587b1d nrfx: cracen: align to nrf54lm20a
=== updating hal_nxp (deps/modules/hal/nxp):
HEAD is now at 7a52cbb7c s32: s32ze: patch PSI5 for integration with zephyr
=== updating hal_stm32 (deps/modules/hal/stm32):
HEAD is now at 1e753266 license: Update Synopsys license terms in the license file and code

~/p/o/io/i/d/zephyr-course > main
west zephyr-export
Zephyr (/home/dms/projects/own/iomico/internal/demo/deps/zephyr/share/zephyr-package/cmake)
has been added to the user package registry in:
~/.cmake/packages/Zephyr

ZephyrUnittest (/home/dms/projects/own/iomico/internal/demo/deps/zephyr/share/zephyrunittest-package/cmake)
has been added to the user package registry in:
~/.cmake/packages/ZephyrUnittest

~/p/o/io/i/d/zephyr-course > main
ls
app build README.md west.yml

~/p/o/io/i/d/zephyr-course > main
```

```
=== updating hal_nxp (deps/modules/hal/nxp):
```

```
HEAD is now at 7a52cbb7c s32: s32ze: patch PSI5 for integration with zephyr
```

```
=== updating hal_stm32 (deps/modules/hal/stm32):
```

```
HEAD is now at 1e753266 license: Update Synopsys license terms in the license file and code
```

```
~/p/o/io/i/d/zephyr-course > main
```

```
✓ < .zephyr ✚ < 19:43:08
```

```
west zephyr-export
```

```
Zephyr (/home/dms/projects/own/iomico/internal/demo/deps/zephyr/share/zephyr-package/cmake)
```

```
has been added to the user package registry in:
```

```
~/.cmake/packages/Zephyr
```

```
ZephyrUnittest (/home/dms/projects/own/iomico/internal/demo/deps/zephyr/share/zephyrunittest-package/cmake)
```

```
has been added to the user package registry in:
```

```
~/.cmake/packages/ZephyrUnittest
```

```
~/p/o/io/i/d/zephyr-course > main
```

```
✓ < .zephyr ✚ < 19:44:09
```

```
ls
```

```
app build README.md west.yml
```

```
~/p/o/io/i/d/zephyr-course > main
```

```
✓ < .zephyr ✚ < 19:56:39
```

```
west build --board frdm_rw612/rw612 app -p
```

```
-- west build: making build dir /home/dms/projects/own/iomico/internal/demo/zephyr-course/build pristine
```

```
-- west build: generating a build system
```

```
Loading Zephyr default modules (Zephyr base).
```

```
-- Application: /home/dms/projects/own/iomico/internal/demo/zephyr-course/app
```

```
-- CMake version: 3.28.3
```

```
-- Found Python3: /home/dms/projects/own/iomico/internal/demo/.zephyr/bin/python3 (found suitable version "3.12.3", minimum required is "3.10") found components: Interpreter
```

```
-- Cache files will be written to: /home/dms/.cache/zephyr
```

```
-- Zephyr version: 4.2.0 (/home/dms/projects/own/iomico/internal/demo/deps/zephyr)
```

```
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
```

```
-- Board: frdm_rw612, qualifiers: rw612
```

```
-- ZEPHYR_TOOLCHAIN_VARIANT not set, trying to locate Zephyr SDK
```

```
-- Found host-tools: zephyr 0.17.4 (/home/dms/zephyr-sdk-0.17.4)
```

```
-- Found toolchain: zephyr 0.17.4 (/home/dms/zephyr-sdk-0.17.4)
```

```
-- Found Dtc: /home/dms/zephyr-sdk-0.17.4/sysroots/x86_64-pokysdk-linux/usr/bin/dtc (found suitable version "1.7.0", minimum required is "1.4.6")
```

```
-- Found BOARD.dts: /home/dms/projects/own/iomico/internal/demo/deps/zephyr/boards/nxp/frdm_rw612/frdm_rw612.dts
```

```
dms@x1G-dev:~/projects/own/iomico/internal/demo/zephyr-course$ main.cpp - demo - Visual Studio Code
Merged configuration '/home/dms/projects/own/iomico/internal/demo/zephyr-course/app/prj.conf'
Configuration saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/.config'
Kconfig header saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLd: /home/dms/zephyr-sdk-0.17.4/arm-zephyr-eabi/arm-zephyr-eabi/bin/ld.bfd (found version "2.38")
-- The C compiler identification is GNU 12.2.0
-- The CXX compiler identification is GNU 12.2.0
-- The ASM compiler identification is GNU
-- Found assembler: /home/dms/zephyr-sdk-0.17.4/arm-zephyr-eabi/bin/arm-zephyr-eabi-gcc
-- Looking for device RW612 in /home/dms/projects/own/iomico/internal/demo/deps/modules/hal/nxp/mcux/mcux-sdk-ng/devices/
-- Found device folder: /home/dms/projects/own/iomico/internal/demo/deps/modules/hal/nxp/mcux/mcux-sdk-ng/devices/Wireless/RW/RW612
-- Found gen_kobject_list: /home/dms/projects/own/iomico/internal/demo/deps/zephyr/scripts/build/gen_kobject_list.py
-- Configuring done (9.6s)
-- Generating done (0.2s)
-- Build files have been written to: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build
-- west build: building application
[1/159] Preparing syscall dependency handling

[2/159] Generating include/generated/zephyr/version.h
-- Zephyr version: 4.2.0 (/home/dms/projects/own/iomico/internal/demo/deps/zephyr), build: v4.2.0
[159/159] Linking CXX executable zephyr/zephyr.elf
Memory region      Used Size  Region Size  %age Used
FLASH:             39324 B      64 MB      0.06%
RAM:               9064 B      960 KB      0.92%
SMU1:              510 KB      510 KB     100.00%
SMU2:              140 KB      140 KB     100.00%
IDT_LIST:           0 GB       32 KB      0.00%
Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: frdm_rw612

~/p/o/io/i/d/zephyr-course > main  ✓ < 27s < .zephyr  < 19:57:36
west flash
-- west flash: rebuilding
ninja: no work to do.
-- west flash: using runner jlink
-- runners.jlink: reset after flashing requested
-- runners.jlink: JLink version: 7.98g
-- runners.jlink: Flashing file: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.bin

~/p/o/io/i/d/zephyr-course > main  ✓ < .zephyr  < 19:58:05
```

Inferior 1 [Remote target] will be killed.

Quit anyway? (y or n) y

```
~/p/o/i/o/i/d/zephyr-course > main ?l
west build -b native_sim app -p
-- west build: making build dir /home/dms/projects/own/iomico/internal/demo/zephyr-course/build pristine
-- west build: generating a build system
Loading Zephyr default modules (Zephyr base).
-- Application: /home/dms/projects/own/iomico/internal/demo/zephyr-course/app
-- CMake version: 3.28.3
-- Found Python3: /home/dms/projects/own/iomico/internal/demo/.zephyr/bin/python3 (found suitable version "3.12.3", minimum required is "3.10") found components: Interpreter
-- Cache files will be written to: /home/dms/.cache/zephyr
-- Zephyr version: 4.2.0 (/home/dms/projects/own/iomico/internal/demo/deps/zephyr)
-- Found west (found suitable version "1.5.0", minimum required is "0.14.0")
-- Board: native_sim, qualifiers: native
-- ZEPHYR_TOOLCHAIN_VARIANT not set, trying to locate Zephyr SDK
-- Found host-tools: zephyr 0.17.4 (/home/dms/zephyr-sdk-0.17.4)
-- Found toolchain: host (gcc/ld)
-- Found Dtc: /home/dms/zephyr-sdk-0.17.4/sysroots/x86_64-pokysdk-linux/usr/bin/dtc (found suitable version "1.7.0", minimum required is "1.4.6")
-- Found BOARD.dts: /home/dms/projects/own/iomico/internal/demo/deps/zephyr/boards/native/native_sim/native_sim.dts
-- Generated zephyr.dts: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.dts
-- Generated pickled edt: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/edt.pickle
-- Generated devicetree_generated.h: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/include/generated/zephyr/devicetree_generated.h
Parsing /home/dms/projects/own/iomico/internal/demo/deps/zephyr/Kconfig
Loaded configuration '/home/dms/projects/own/iomico/internal/demo/deps/zephyr/boards/native/native_sim/native_sim_defconfig'
Merged configuration '/home/dms/projects/own/iomico/internal/demo/zephyr-course/app/prj.conf'
Configuration saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/.config'
Kconfig header saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLd: /usr/bin/ld.bfd (found version "2.42")
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- The ASM compiler identification is GNU
-- Found assembler: /usr/bin/gcc
-- Found gen_kobject_list: /home/dms/projects/own/iomico/internal/demo/deps/zephyr/scripts/build/gen_kobject_list.py
```

✓ < 1m 55s < .zephyr < 20:03:39

```
dms@x1G-dev:~/projects/own/iomico/internal/demo/zephyr-course | main.cpp - demo - Visual Studio Code
Configuration saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/.config'
Kconfig header saved to '/home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/include/generated/zephyr/autoconf.h'
-- Found GnuLd: /usr/bin/ld.bfd (found version "2.42")
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- The ASM compiler identification is GNU
-- Found assembler: /usr/bin/gcc
-- Found gen_kobject_list: /home/dms/projects/own/iomico/internal/demo/deps/zephyr/scripts/build/gen_kobject_list.py
-- Configuring done (8.9s)
-- Generating done (0.1s)
-- Build files have been written to: /home/dms/projects/own/iomico/internal/demo/zephyr-course/build
-- west build: building application
[1/96] Preparing syscall dependency handling

[2/96] Generating include/generated/zephyr/version.h
-- Zephyr version: 4.2.0 (/home/dms/projects/own/iomico/internal/demo/deps/zephyr), build: v4.2.0
[94/96] Linking CXX executable zephyr/zephyr.elf
Generating files from /home/dms/projects/own/iomico/internal/demo/zephyr-course/build/zephyr/zephyr.elf for board: native_sim
[96/96] Running utility command for native_runner_executable

~/p/o/io/i/d/zephyr-course > main ?1  ✓ < 12s < .zephyr  < 20:04:10
west flash
-- west flash: rebuilding
[2/2] Running utility command for native_runner_executable
-- west flash: using runner native
*** Booting Zephyr OS build v4.2.0 ***
[00:00:00.000,000] <inf> main: LED state: OFF
[00:00:01.010,000] <inf> main: LED state: ON
[00:00:02.020,000] <inf> main: LED state: OFF
[00:00:03.030,000] <inf> main: LED state: ON
[00:00:04.040,000] <inf> main: LED state: OFF
[00:00:05.050,000] <inf> main: LED state: ON
[00:00:06.060,000] <inf> main: LED state: OFF
^C
Stopped at 6.170s

~/p/o/io/i/d/zephyr-course > main ?1  INT ✗ < 7s < .zephyr  < 20:04:24
```



Troubleshooting



Common Issues



Problem	Cause	Fix
<code>find_package(Zephyr)</code> fails	Missing env setup	Run <code>west zephyr-export</code>
Build fails with missing board	Wrong board name	Check <code>west boards</code>
Missing HAL errors	Incomplete manifest	Add module to <code>name-allowlist</code>
SDK version mismatch	Wrong SDK for Zephyr version	Check compatibility table



SDK Version Compatibility



- Each Zephyr release specifies a **minimum SDK version**
- Check: [Zephyr SDK Toolchain docs](#)
- When upgrading Zephyr, always verify SDK compatibility
- SDK can usually be updated independently



Tasks



0. **Read:** [West documentation](#)
1. **Fork and set up the provided workspace** — <https://github.com/iomicro-public/zephyr-course>
2. **Build and run an LED blink application**
 - Flash to your hardware board
 - Verify the LED toggles every second (if there's an LED)
 - Push tag: `l2-task1`



Homework (cont.)



3. (Optional) **Build and run on** `native_sim`

- Check the boot header for the OS version

```
*** Booting Zephyr OS build v4.2.0 ***
```

Thank You!





Resources



1. [West \(Zephyr's meta-tool\) documentation](#)
2. [West manifest format](#)
3. [Zephyr application development](#)
4. [Zephyr SDK Toolchain docs](#)
5. [Supported boards](#)